PLM Intelligence

Turning a raw OEM XML extract into traceable decisions for an e-motor supplier: explore the product, identify risks, assess the impact of a change.

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1 PROBLEM STATEMENT

We play the engineering team of an electric motor supplier. Our OEM customer develops the vehicle in 3DEXPERIENCE, and we have no access to that environment. All we get is an XML extract of the product: a **410 kW front e-drive, 100.1 kg, 79 products, 36 physical parts** and three subassemblies (e-motor, reducer, housing).

The file holds masses, materials and revisions. It has **no cost, no supplier and no quality data**, one revision per part and everything still IN_WORK. We still have to make cost and risk decisions, and defend every figure in front of the OEM.

Read naively, the file gives wrong answers. A part and its envelope (wrapper) carry the same mass, so summing both counts it twice. Four products share the same name. Two stator meshing objects are not real parts.

So the real requirement is **traceability**: any number we show must link back to the XML element, or the external data row, it was computed from.

Understand
Navigate the BOM, inspect parts, versions and physical properties.

Identify risks
Which parts are critical? Where do we depend on one supplier?

Assess a change
If a part changes, which assemblies, suppliers and costs move?

2 APPROACH AND METHODOLOGY

- **Traceability first.** Every object read from the XML keeps its file, line and element. Every computed value (cost, risk score, impact, AI answer) carries the sources behind it.
- **One shared data contract** (keys, units, cost rules) so five workstreams could build in parallel: data, analysis, interface, AI assistant, pitch.
- **Filling the gaps, openly.** We wrote 14 material prices, 15 suppliers and 53 sourcing rows (supplier, make or buy, lead time). Each row states its source. Copper and aluminium follow LME prices; the rest are flagged as estimates or assumptions.
- **Explicit rules.** Material cost = mass × price per kg, on leaf parts only; summed over the 36 parts it gives back exactly 100.1 kg. Risk = probability × impact over six criteria scored 0 to 1: single source, supplier concentration, maturity; cost share, lead time, reuse. Price risk is scored apart from supply risk.
- **A constrained AI.** The language model never reads the file and does no arithmetic. It calls 14 tools that return computed facts, each with an evidence id. Invented citations are removed, and an answer without evidence is replaced by a scope message. We picked the model by testing three candidates on trap questions.
- **Validation.** 282 automated tests, and an automatic evaluation of the assistant on 17 reference questions.

3 THE PROPOSED SOLUTION

A web application running on the real extract.

OEM XML + team data › **Data model** with sources › **Cost · Risk · Change impact** › **Web app** + AI assistant

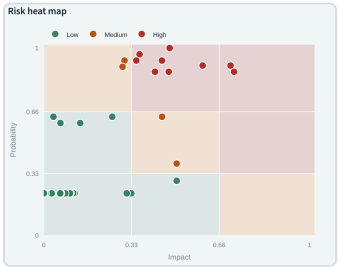
- **Understand:** collapsible BOM tree, part card with its XML line, comparison of parts, revisions and maturity.
- **3D model of the motor:** the app shows the motor in 3D so you can see in real time where each part sits. The selected part is highlighted, parts are coloured by risk level, and a changed component is shown inside its subassembly. Positions come from the XML.
- **Identify risks:** part ranking and heat map, the six criteria explained per part, exposure per supplier, material and raw material price.
- **Assess a change:** change a part's mass or material, or swap it for another part, and see the impacted assemblies up to the root, the suppliers, lead times, material cost and risk before and after.
- **AI assistant:** questions in English or French; every answer cites the XML elements and data rows it rests on.
- Not tied to this motor: any XPDMXML extract in the same format can be loaded. Missing data is reported as missing, never shown as zero.

4 RESULTS

26%
of the material cost stops if Nemak fails: sole source of 5 housing parts, 112-day lead time

35%
of the material cost is copper, for 7% of the mass: the most volatile material

57%
of the material cost sits in the electric motor, for 40% of the mass



9 of the 34 scored parts are high risk. The three most critical are Nemak housing castings, all single-sourced with a 112-day lead time. GKN Driveline delivers 8 parts, 6 of them single-sourced. The housing holds 5 of the 9 high-risk parts.

Change impact in one screen. Making the differential side shaft 20% lighter touches 5 assemblies up to the root (it is used twice), 2 suppliers, and saves 0.36 kg.

Left: risk heat map of the 34 scored parts, taken from the tool (probability × impact).

Limits we state openly. Prices, suppliers and lead times are team assumptions, labelled as such in the tool: the scores are only as good as these inputs, the method holds with real purchasing data. Costs are material costs only (no purchase, labour or tooling). The extract has no quality data and no version history. 3D shapes are illustrative; their positions come from the XML. **Next steps:** real ERP data, full cost and successive extracts for history, then a direct 3DEXPERIENCE connection with an internally hosted AI model.

Code: github.com/quentin-trlt/hackathon-plm-intelligence-dassault-systems